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REVIEW ARTICLE

A Review- Biomedical Engineering-Present and Future Prospective

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ABSTRACT:
Biomedical engineering is the application of engineering principles and design concepts to medicine and biology for healthcare purposes. This field seeks to close the gap between engineering and medicine: It combines the design and problem solving skills of engineering with medical and biological sciences to advance healthcare treatment, including diagnosis, monitoring, and therapy. Much of the work in biomedical engineering consists of research and development, spanning a broad array of subfields. Prominent biomedical engineering applications include the development of biocompatible prostheses, various diagnostic and therapeutic medical devices ranging from clinical equipment to micro-implants, common imaging equipment such as MRIs and EEGs, regenerative tissue growth, pharmaceutical drugs and therapeutic biological. But more often, sub-disciplines within BME are classified by their association(s) with other more established engineering fields, which can include:

- Biochemical-BME, based on Chemical engineering - often associated with biochemical, cellular, molecular and tissue engineering, biomaterials, and biotransport.
- Bioelectrical-BME, based on Electrical engineering and Computer Science - often associated with bioelectrical and neural engineering, bioinstrumentation, biomedical imaging, and medical devices. This also tends to encompass optics and optical engineering - biomedical optics, bioinformatics, imaging and related medical devices.
- Biomechanical-BME, based on Mechanical engineering - often associated with biomechanics, biotransport, medical devices, and modeling of biological systems, like soft tissue mechanics.

RoHS seeks to limit the dangerous substances in circulation in electronics products, in particular toxins and heavy metals, which are subsequently released into the environment when such devices are recycled. IEC 60601-1-11 (2010) must now be incorporated into the design and verification of a wide range of home use and point of care medical devices along with other applicable standards in the IEC 60601 3rd edition series.

KEYWORDS: Diagnosis, Monitoring, Therapy, Biocompatible prostheses, RoHS, IEC 60601-1-11.

INTRODUCTION:
Biomedical engineering is the application of engineering principles and design concepts to medicine and biology for healthcare purposes. This field seeks to close the gap between engineering and medicine: It combines the design and problem solving skills of engineering with medical and biological sciences to advance healthcare treatment, including diagnosis, monitoring, and therapy.

Biomedical engineering is a discipline that advances knowledge in engineering, biology and medicine, and improves human health through cross-disciplinary activities that integrate the engineering sciences with the biomedical sciences and clinical practice. It includes:

1. The acquisition of new knowledge and understanding of living systems through the innovative and substantive application of experimental and analytical techniques based on the engineering sciences.
2. The development of new devices, algorithms, processes and systems that advance biology and medicine and improve medical practice and health care delivery.

Biomedical engineering has only recently emerged as its own discipline, compared to many other engineering fields.

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Such an evolution is common as a new field transitions from being an interdisciplinary specialization among already-established fields, to being considered a field in itself. Much of the work in biomedical engineering consists of research and development, spanning a broad array of subfields. Prominent biomedical engineering applications include the development of biocompatible prostheses, various diagnostic and therapeutic medical devices ranging from clinical equipment to micro-implants, common imaging equipment such as MRIs and EEGs, regenerative tissue growth, pharmaceutical drugs and therapeutic biologicals.[1]

Development of Bioengineering:

Over the last few years there has been a major paradigm shift in both Europe and the United States away from traditional schemes of health care towards health care systems which are much more dependent on technology. This is true in terms of diagnosis (eg body scanners); treatment (radiation therapy and minimal access surgery); and health care system integration (via information technology). In parallel with these changes, there has been a progressive increase in the proportion of the national Gross Domestic Product spent in the medical sector. For example, in the United Kingdom it is currently between 6 and 7%, in Germany about 9%, and in the United States about 14%. This has resulted partly from demographic changes and additionally from increasing public demand for better health care. As medical practice becomes more technologically based, a progressive shift is occurring in industry to meet the demand. Developments in science and engineering are increasingly being directed away from traditional technologies towards those required for health care in its widest sense. Although in many countries there is a problem with escalating costs in the medical sector, technology can contribute to economies because of falling costs of electronic/physics based components relative to those for personnel, and because of technologically based screening programmes.[2]

Where do they Work?

Biomedical engineers are employed in industry, in hospitals, in research facilities of educational and medical institutions, in teaching, and in government regulatory agencies. They often serve a coordinating or interfacing function, using their background in both the engineering and medical fields. In industry, they may create designs where an in-depth understanding of living systems and of technology is essential. They may be involved in performance testing of new or proposed products. Government positions often involve product testing and safety, as well as establishing safety standards for devices. In the hospital, the biomedical engineer may provide advice on the selection and use of medical equipment, as well as supervising its performance testing and maintenance[3]. They may also build customized devices for special health care or research needs. In research institutions, biomedical engineers supervise laboratories and equipment, and participate in or direct research activities in collaboration with other researchers with such backgrounds as medicine, physiology, and nursing. Some biomedical engineers are technical advisors for marketing departments of companies and some are in management positions. Some biomedical engineers also have advanced training in other fields. For example, many biomedical engineers also have an M.D. degree, thereby combining an understanding of advanced technology with direct patient care or clinical research.

Notable subdisciplines within biomedical engineering:

Biomedical engineering can be viewed from two angles, from the medical applications side and from the engineering side. A biomedical engineer must have some view of both sides[4]. As with many medical specialties (e.g. cardiology, neurology), some BME sub-disciplines are identified by their associations with particular systems of the human body, such as:

- **Cardiovascular technology** - which includes all drugs, biologics, and devices related with diagnostics and therapeutics of cardiovascular systems
- **Neural technology** - which includes all drugs, biologics, and devices related with diagnostics and therapeutics of the brain and nervous systems
- **Orthopaedic technology** - which includes all drugs, biologics, and devices related with diagnostics and therapeutics of skeletal systems
- **Cancer technology** - which includes all drugs, biologics, and devices related with diagnostics and therapeutics of cancer

But more often, sub-disciplines within BME are classified by their association(s) with other more established engineering fields, which can include[5]:

- **Biochemical-BME, based on Chemical engineering** - often associated with biochemical, cellular, molecular and tissue engineering, biomaterials, and biotransport.
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One more way to sub-classify the discipline is on the basis of the products created. [6]

Biologics and Biopharmaceuticals often designed using the principles of synthetic biology (synthetic biology is an extension of genetic engineering). The design of biologic and biopharma products comes broadly under the BME-related (and overlapping) disciplines of biotechnology and bioengineering.

Pharmaceutical Drugs (so-called "small-molecule" or non-biologic) which are commonly designed using the principles of synthetic chemistry and traditionally discovered using high-throughput screening methods at the beginning of the development process.

Tissue engineering

Tissue engineering, like genetic engineering, is a major segment of Biotechnology - which overlaps significantly with BME .One of the goals of tissue engineering is to create artificial organs (via biological material) for patients that need organ transplants. Biomedical engineers are currently researching methods of creating such organs. Researchers have grown solid jawbones and tracheas from human stem cells towards this end. Several artificial urinary bladders actually have been grown in laboratories and transplanted successfully into human patients. Bioartificial organs, which use both synthetic and biological components, are also a focus area in research, such as with hepatic assist devices that use liver cells within an artificial bioreactor construct.[7]

Genetic engineering

Genetic engineering, recombinant DNA technology, genetic modification/manipulation (GM) and gene splicing are terms that apply to the direct manipulation of an organism's genes. Genetic engineering is different from traditional breeding, where the organism's genes are manipulated indirectly. Genetic engineering uses the techniques of molecular cloning and transformation to alter the structure and characteristics of genes directly. Genetic engineering techniques have found success in numerous applications.

Neural engineering

Neural engineering (also known as Neuroengineering) is a discipline that uses engineering techniques to understand, repair, replace, or enhance neural systems. Neural engineers are uniquely qualified to solve design problems at the interface of living neural tissue and non-living constructs.[8]

Pharmaceutical engineering

Pharmaceutical engineering is sometimes regarded as a branch of biomedical engineering, and sometimes a branch of chemical engineering; in practice, it is very much a hybrid sub-discipline .Aside from those pharmaceutical products directly incorporating biological agents or materials, even developing chemical drugs is considered to require substantial BME knowledge due to the physiological interactions inherent to such products' usage. With the increasing prevalence of "combination products," the lines are now blurring among healthcare products such as drugs, biologics, and various types of devices.

Medical devices

This is an extremely broad category—essentially covering all health care products that do not achieve their intended results through predominantly chemical (e.g., pharmaceuticals) or biological (e.g., vaccines) means, and do not involve metabolism.[9]

A medical device is intended for use in:

- The diagnosis of disease or other conditions, or
- In the cure, mitigation, treatment, or prevention of disease

Some examples include pacemakers, infusion pumps, the heart-lung machine, dialysis machines, artificial organs, implants, artificial limbs, corrective lenses, cochlear implants, ocular prosthetics, facial prosthetics, somato prosthetics, and dental implants.

Medical devices are regulated and classified (in the US) as follows:

1. **Class I devices** present minimal potential for harm to the user and are often simpler in design than Class II or Class III devices. Devices in this category include tongue depressors, bedpans, elastic bandages, examination gloves, and hand-held surgical instruments and other similar types of common equipment.
2. **Class II devices** are subject to special controls in addition to the general controls of Class I devices. Special controls may include special labeling requirements, mandatory performance standards, and postmarket surveillance. Devices in this class are typically non-invasive and include x-ray machines, PACS, powered wheelchairs, infusion pumps, and surgical drapes.
3. **Class III devices** generally require premarket approval (PMA) or premarket notification (510k), a scientific review to ensure the device's safety and effectiveness, in addition to the general controls of Class I. Examples include replacement heart valves, hip and knee joint implants, silicone gel-filled breast implants, implanted cerebellar stimulators, implantable pacemaker pulse generators and endosseous (intra-bone) implants.

Medical imaging

Medical/biomedical imaging is a major segment of medical devices. This area deals with enabling clinicians to directly or indirectly "view" things not visible in plain sight (such as due to their size, and/or location). This can involve utilizing ultrasound, magnetism, UV, other radiology, and other means.

Imaging technologies are often essential to medical diagnosis, and are typically the most complex equipment found in a hospital including: [10]

- Fluoroscopy
- Magnetic resonance imaging (MRI)
- Nuclear medicine
- Positron emission tomography (PET) scans PET
- Projection radiography such as X-rays and CT scans
- Tomography
- Ultrasound
- Optical microscopy
- Electron microscopy

Implants
An implant is a kind of medical device made to replace and act as a missing biological structure. The surface of implants that contact the body might be made of a biomedical material such as titanium, silicone or apatite depending on what is the most functional. In some cases implants contain electronics e.g. artificial pacemaker and cochlear implants. Some implants are bioactive, such as subcutaneous drug delivery devices in the form of implantable pills or drug-eluting stents.[11]

Bionics
Artificial body part replacement is just one of the things that bionics can do. Concerned with the intricate and thorough study of the properties and function of human body systems, bionics may be applied to solve some engineering problems. Careful study of the different function and processes of the eyes, ears, and other organs paved the way for improved cameras, television, radio transmitters and receivers, and many other useful tools. These developments have indeed made our lives better, but the best contribution that bionics has made is in the field of biomedical engineering.

Clinical engineering
Clinical engineering is the branch of biomedical engineering dealing with the actual implementation of medical equipment and technologies in hospitals or other clinical settings. Major roles of clinical engineers include training and supervising biomedical equipment technicians (BMETs), selecting technological products/services and logistically managing their implementation, working with governmental regulators on inspections/audits, and serving as technological consultants for other hospital staff. Clinical engineers also advise and collaborate with medical device producers regarding prospective design improvements based on clinical experiences, as well as monitor the progression of the state-of-the-art so as to redirect procurement patterns accordingly.[12]

Regulatory issues
Regulatory issues are of particular concern to a biomedical engineer; it is among the most heavily-regulated fields of engineering, and practicing biomedical engineers must routinely consult and cooperate with regulatory law attorneys and other experts. The Food and Drug Administration (FDA) is the principal healthcare regulatory authority in the United States, having jurisdiction over medical devices, drugs, biologics, and combination products. The paramount objectives driving policy decisions by the FDA are safety and efficacy of healthcare products.

The different regulatory arrangements sometimes result in particular technologies being developed first for either the U.S. or in Europe depending on the more favorable form of regulation. While nations often strive for substantive harmony to facilitate cross-national distribution, philosophical differences about the optimal extent of regulation can be a hindrance; more restrictive regulations seem appealing on an intuitive level, but critics decry the tradeoff cost in terms of slowing access to life-saving developments.[13]

RoHS II
Directive 2011/65/EU, better known as RoHS 2 is a recast of legislation originally introduced in 2002. The original EU legislation "Restrictions of Certain Hazardous Substances in Electrical and Electronics Devices" (RoHS Directive 2002/95/EC) was replaced and superseded by 2011/65/EU published in July 2011 and commonly known as RoHS 2. RoHS seeks to limit the dangerous substances in circulation in electronics products, in particular toxins and heavy metals, which are subsequently released into the environment when such devices are recycled.

The scope of RoHS 2 is widened to include products previously excluded, such as medical devices and industrial equipment. In addition, manufacturers are now obliged to provide conformity risk assessments and test reports – or explain why they are lacking. For the first time, not only manufacturers, but also importers and distributors share a responsibility to ensure Electrical and Electronic Equipment within the scope of RoHS comply with the hazardous substances limits and have a CE mark on their products.[14]

IEC 60601
The new International Standard IEC 60601 for home healthcare electro-medical devices defining the requirements for devices used in the home healthcare environment. IEC 60601-1-11 (2010) must now be incorporated into the design and verification of a wide range of home use and point of care medical devices along with other applicable standards in the IEC 60601 3rd edition series.

The mandatory date for implementation of the EN European version of the standard is June 1, 2013. The US FDA requires the use of the standard on June 30, 2013, while Health Canada recently extended the required date from June 2012 to April 2013. The North American agencies will only require these standards for new device submissions, while the EU will take the more severe approach of requiring all applicable devices being placed on the market to consider the home healthcare standard.[15]

Founding figures[16]

- Leslie Geddes (deceased)- Professor Emeritus at Purdue University, electrical engineer, inventor, and educator of over 2000 biomedical engineers, received a National Medal of Technology in 2006 from President George Bush for his more than 50 years of contributions that have spawned innovations ranging from burn treatments to miniature defibrillators, ligament repair to tiny blood pressure monitors for premature infants, as well as a new method for performing cardiopulmonary resuscitation

(CPR).

- Y. C. Fung - professor emeritus at the University of California, San Diego, considered by many to be the founder of modern Biomechanics
- Robert Langer - Institute Professor at MIT, runs the largest BME laboratory in the world, pioneer in drug delivery and tissue engineering
- Herbert Lissner (deceased) - Professor of Engineering Mechanics at Wayne State University. Initiated studies on blunt head trauma and injury thresholds beginning in 1939 in collaboration with Dr. E.S. Gurdjian, a neurosurgeon at Wayne State's School of Medicine. Individual for whom the American Society of Mechanical Engineers' top award in Biomedical Engineering, the Herbert R. Lissner Medal, is named.
- Nicholas A. Peppas - Chaired Professor in Engineering, University of Texas at Austin, pioneer in drug delivery, biomaterials, hydrogels and nanobiotechnology.
- Otto Schmitt (deceased) - biophysicist with significant contributions to BME, working with biomimetics
- Ascher Shapiro (deceased) - Institute Professor at MIT, contributed to the development of the BME field, medical devices (e.g. intra-aortic balloons)
- John G. Webster - Professor Emeritus at the University of Wisconsin–Madison, a pioneer in the field of instrumentation amplifiers for the recording of electrophysiological signals
- Robert Plonsey - Professor Emeritus at Duke University, pioneer of electrophysiology
- U. A. Whitaker (deceased) - provider of The Whitaker Foundation, which supported research and education in BME by providing over $700 million to various universities, helping to create 30 BME programs and helping finance the construction of 13 buildings
- Frederick Thurstone (deceased) - Professor Emeritus at Duke University, pioneer of diagnostic ultrasound
- Kenneth R. Diller - Chaired and Endowed Professor in Engineering, University of Texas at Austin. Founded the BME department at UT Austin. Pioneer in bioheat transfer, mass transfer, and biotransport
- Alfred E. Mann - Physicist, entrepreneur and philanthropist. A pioneer in the field of Biomedical Engineering.
- Forrest Bird - aviator and pioneer in the invention of mechanical ventilators
- Willem Johan Kolff (deceased) - pioneer of hemodialysis as well as in the field of artificial organs
- John James Rickard Macleod (deceased) - one of the co-discoverers of insulin at Case Western Reserve University.

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